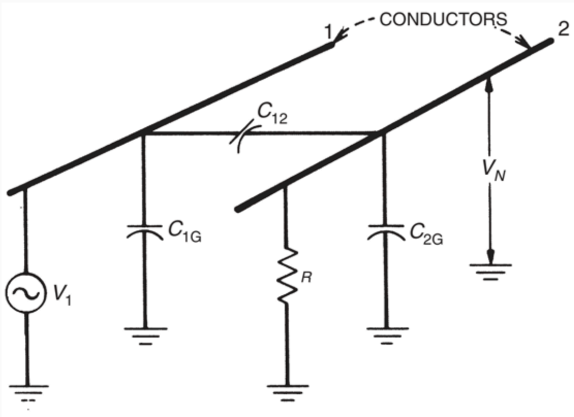
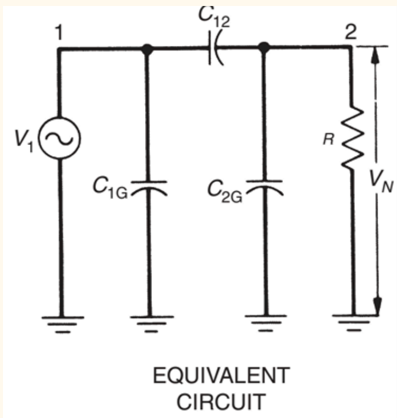
Capacitive Coupling

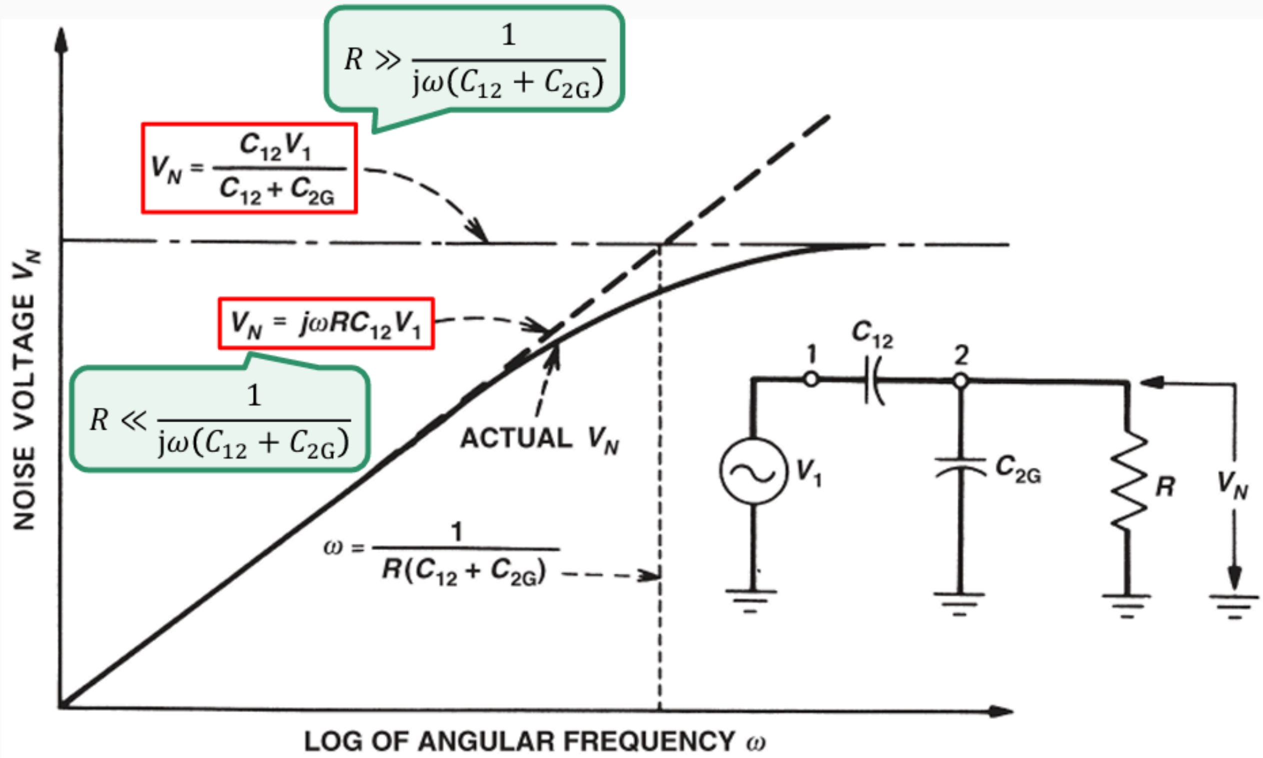




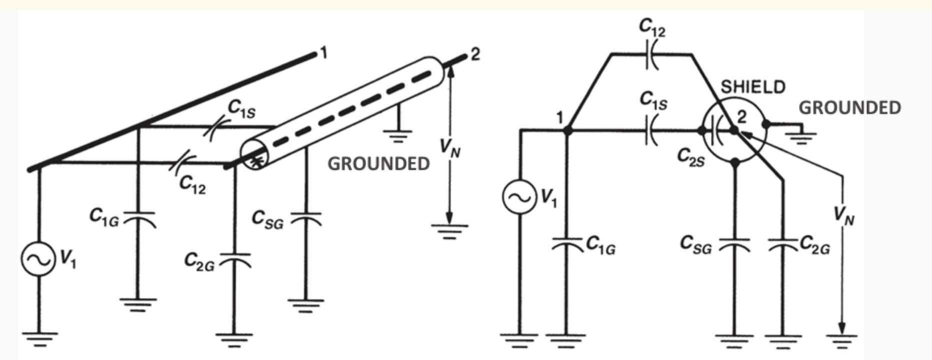
$$V_N = \frac{R||\frac{1}{j\omega C_{2G}}}{\frac{1}{j\omega C_{12}} + \left(R||\frac{1}{j\omega C_{2G}}\right)} V_1$$

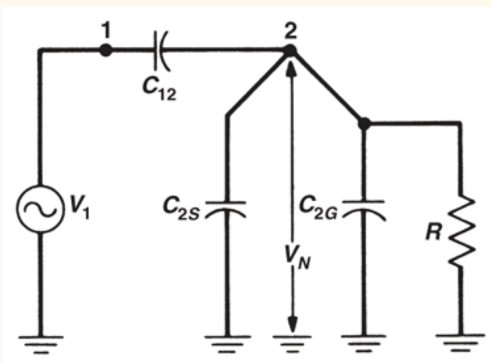
$$V_N = \frac{j\omega C_{12} R}{1 + j\omega(C_{12} + C_{2G})R} V_1$$

V_N when $R \gg \frac{1}{\omega(C_{12}+C_{2G})}$ has a larger magnitude than V_N when $R \ll \frac{1}{\omega(C_{12}+C_{2G})}$.



Electric Field Shielding



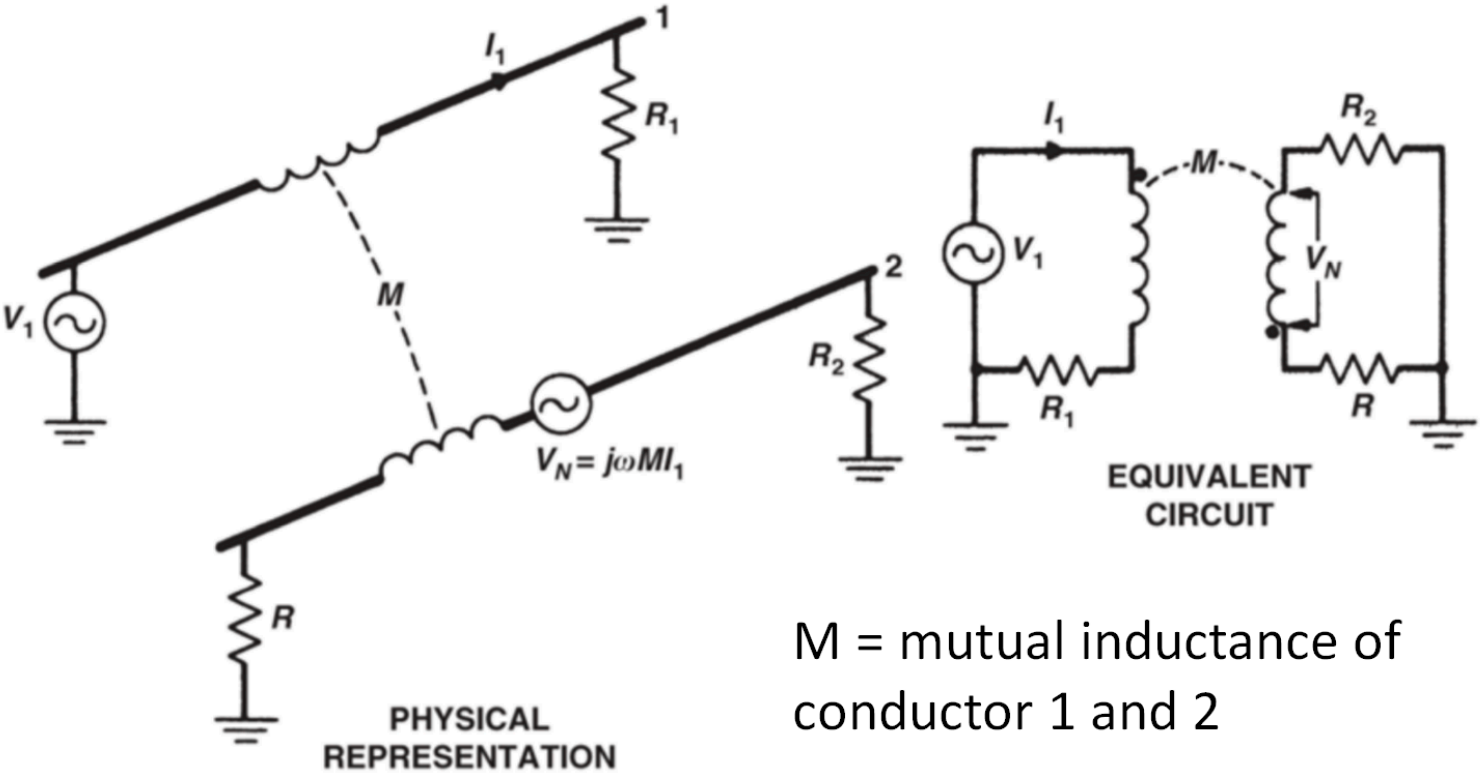


- Shield must be grounded. If conductor and shield are both floating VN = VS

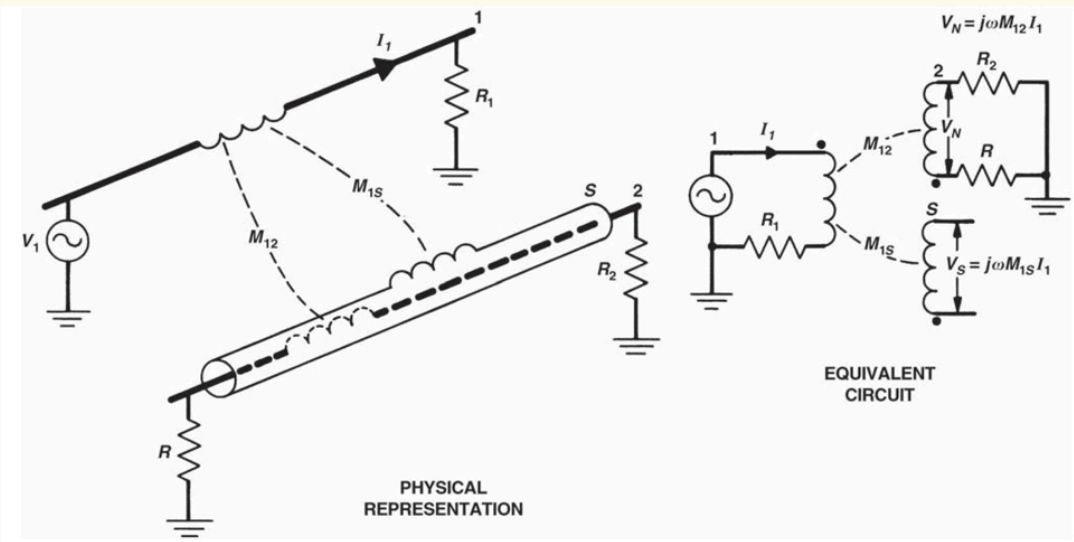
When $R \ll \dfrac{1}{\omega(C_{12}+C_{2G}+C_{2S})}$: $\quad V_N = \dfrac{j\omega C_{12}R}{1+j\omega(C_{12}+C_{2G}+C_{2S})R}V_1 \quad \Rightarrow \quad \boxed{V_N = j\omega C_{12} R V_1}$

When $R \gg \dfrac{1}{\omega(C_{12}+C_{2G}+C_{2S})}$: $\quad V_N = \dfrac{j\omega C_{12}R}{1+j\omega(C_{12}+C_{2G}+C_{2S})R}V_1 \quad \Rightarrow \quad \boxed{V_N = \dfrac{C_{12}}{C_{12}+C_{2G}+C_{2S}}V_1}$

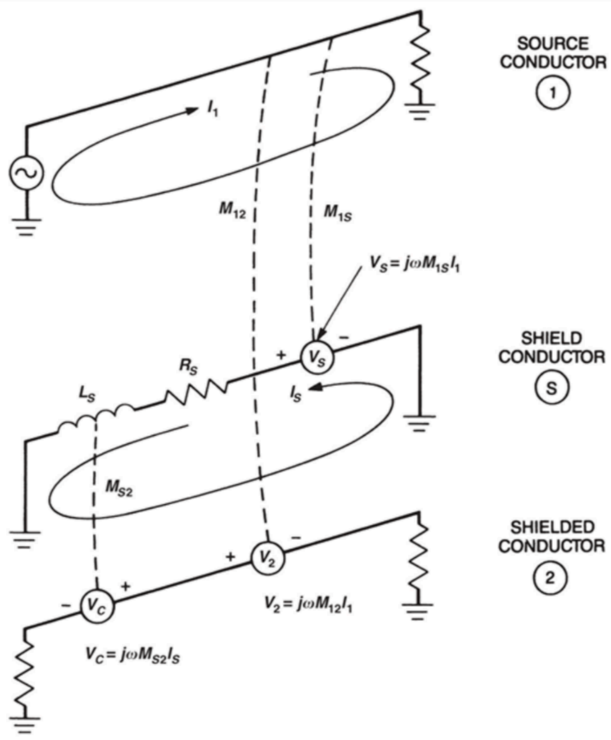
Inductive Coupling



Magnetic Field Shielding



- shield must be grounded at both ends for shield current to flow and act as shield



There are two voltage noises induced on conductor 2:

- V_2 is induced directly from conductor 1
- V_C is induced from the induced shield current

V_2 and V_C have different polarities, hence

$$\boxed{V_N = V_2 - V_C}$$

$$V_2 = \mathrm{j}\omega M_{12} I_1 \qquad V_S = \mathrm{j}\omega M_{1S} I_1$$

$$V_C = \frac{\mathrm{j}\omega}{\mathrm{j}\omega + \omega_c} V_S \qquad M_{1S} = M_{12}$$

omega_c = Rs/Ls

$$\boxed{V_N = \mathrm{j}\omega M_{12} I_1 \left(\frac{\omega_c}{\mathrm{j}\omega + \omega_c}\right)}$$

Low frequency ($\omega \ll \omega_c$)

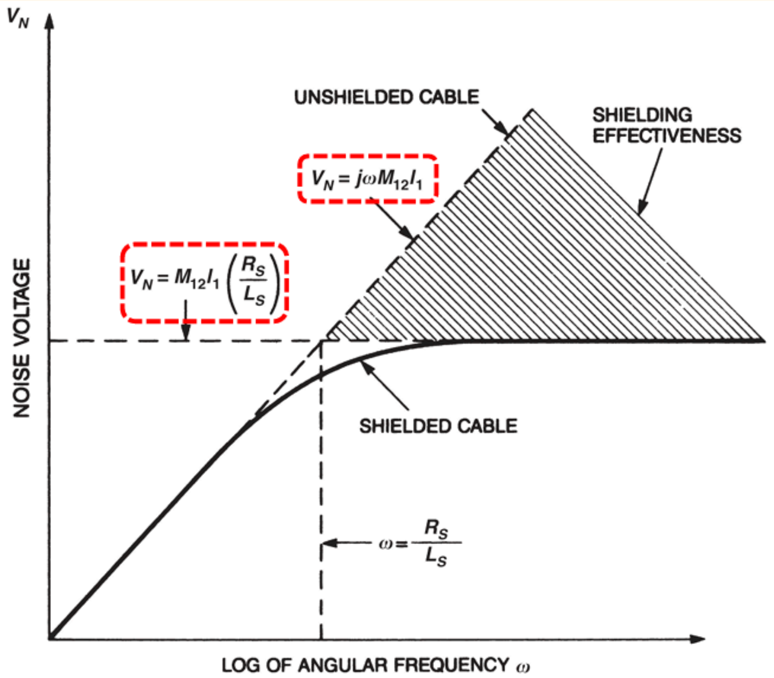
$$V_N = \mathrm{j}\omega M_{12} I_1$$

(Same as w/o shield)

High frequency ($\omega \gg \omega_c$)

$$V_N = M_{12} I_1 \omega_c$$

The shield does not help at low frequencies but limits the inductive noise at high frequencies.



$$\boxed{V_N = \mathrm{j}\omega M_{12} I_1 \left(\frac{R_S/L_S}{\mathrm{j}\omega + R_S/L_S}\right)}$$

- ❑ To minimize pickup noise, one can reduce the shield resistance R_S.
- ❑ It is the induced shield current I_S that produces the magnetic field, which cancels a large percentage of the direct induction into conductor 2 from conductor 1.

In the ideal case, when the shield is short circuit, V_N is fully cancelled.